

Hamilton + KEGG-prior Guild Model

Progress Report — Oral Microbiome Community Dynamics

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Background & Motivation

Research question: Can metabolomic network constraints improve community dynamics modelling?

Existing models

Model	LOO-CV RMSE
gLV (free)	0.0588
gLV + KEGG	0.0741
Hamilton (free)	0.0855
Hamilton + KEGG	← this work

Approach

- **10-guild** replicator (class-level taxonomy)
- **8 patients**, 3 time points (weeks 1 → 2 → 3)
- Sign prior derived from **KEGG/HMDB** metabolite flow matrix F
- Fit via JAX autodiff + L-BFGS-B on RTX 3090

Hypothesis: KEGG sign constraints regularize A towards biologically meaningful interactions without sacrificing fit quality.

Method

Hamilton replicator ODE

$$\dot{\phi}_i = \phi_i \left(b_i + \sum_j A_{ij} \phi_j - \bar{f}(\phi) \right), \quad \bar{f} = \sum_i \phi_i f_i$$

Loss function with metabolite sign penalty

$$\mathcal{L} = \underbrace{\text{RMSE}(\hat{\phi}, \phi)}_{\text{data fit}} + \underbrace{\sum_{(i,j): F_{ij} \neq 0} \frac{|F_{ij}|}{2\sigma^2} \max(0, -\text{sgn}(F_{ij}) \cdot A_{ij})^2}_{\text{metabolite sign penalty } (\sigma=0.15)} + \lambda \|A\|^2$$

Sign prior — data source

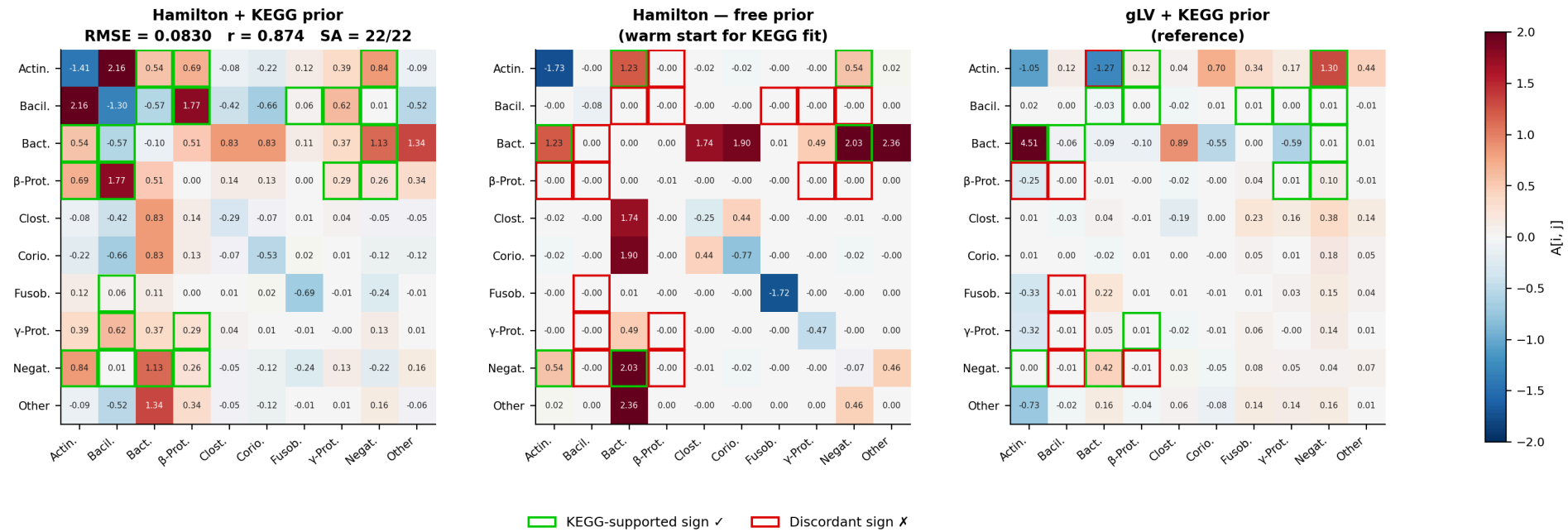
- **Szafranski et al. Suppl. File 1** (literature-curated microbe × metabolite interactions: PRODUCES / USES / IS_INHIBITED_BY)
- KEGG compound ID / HMDB ID → **confidence weight only** (2.0 if annotated, 1.0 otherwise); no live API query
- $F[i, j] > 0$: guild j produces what i uses → expect $A_{ij} > 0$

Optimization

- Warm start from Hamilton free fit
- JAX `jax.vmap` over patients
- scipy L-BFGS-B + analytical gradients
- Diagonal $A_{ii} \leq 0$ (self-limitation)
- 22 sign-constrained pairs (after symmetrisation)

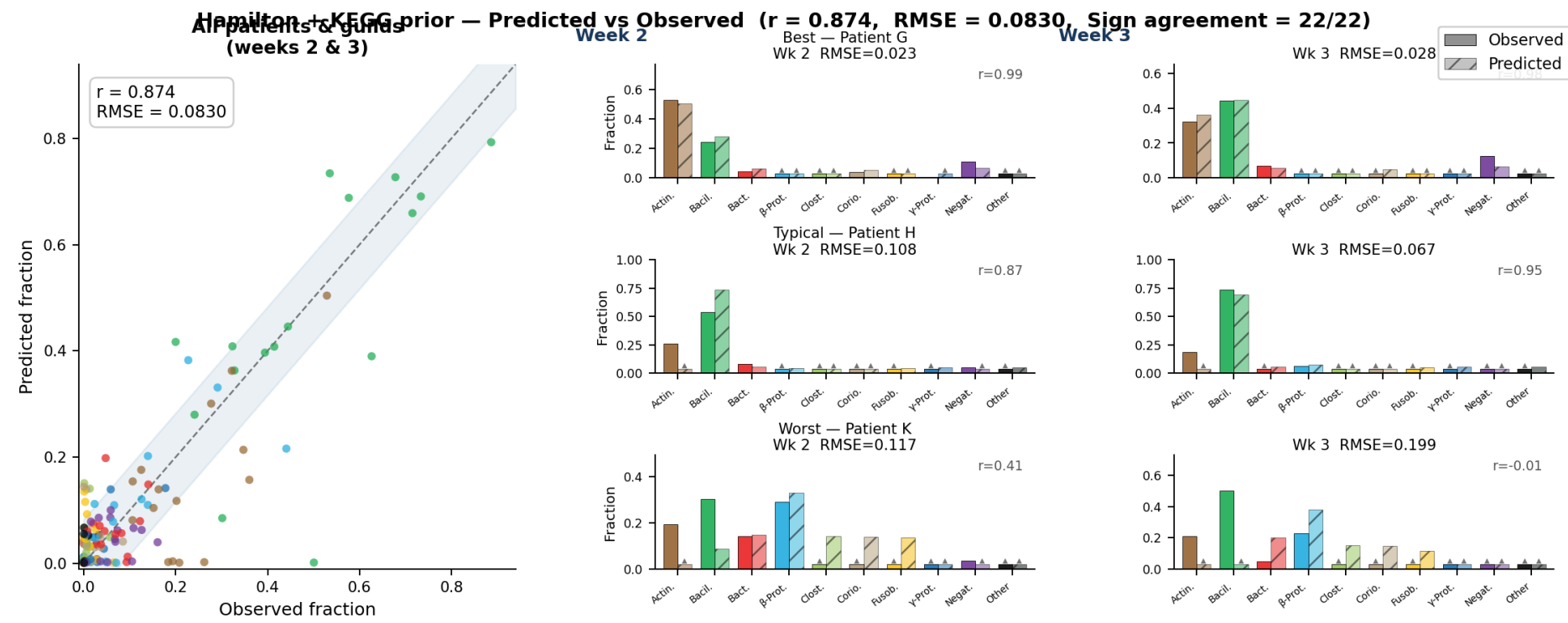
Result 1 — Interaction Matrix A

Guild-level interaction matrices — Hamilton KEGG-prior vs baselines



Sign agreement = 22 / 22 (100%) with KEGG metabolite network · Green border ✓ = supported · Red ✗ = discordant

Result 2 — Trajectory Prediction (Week 1 → 2 → 3)



0.874

Pearson r (overall)

0.0830

RMSE (overall)

0.987

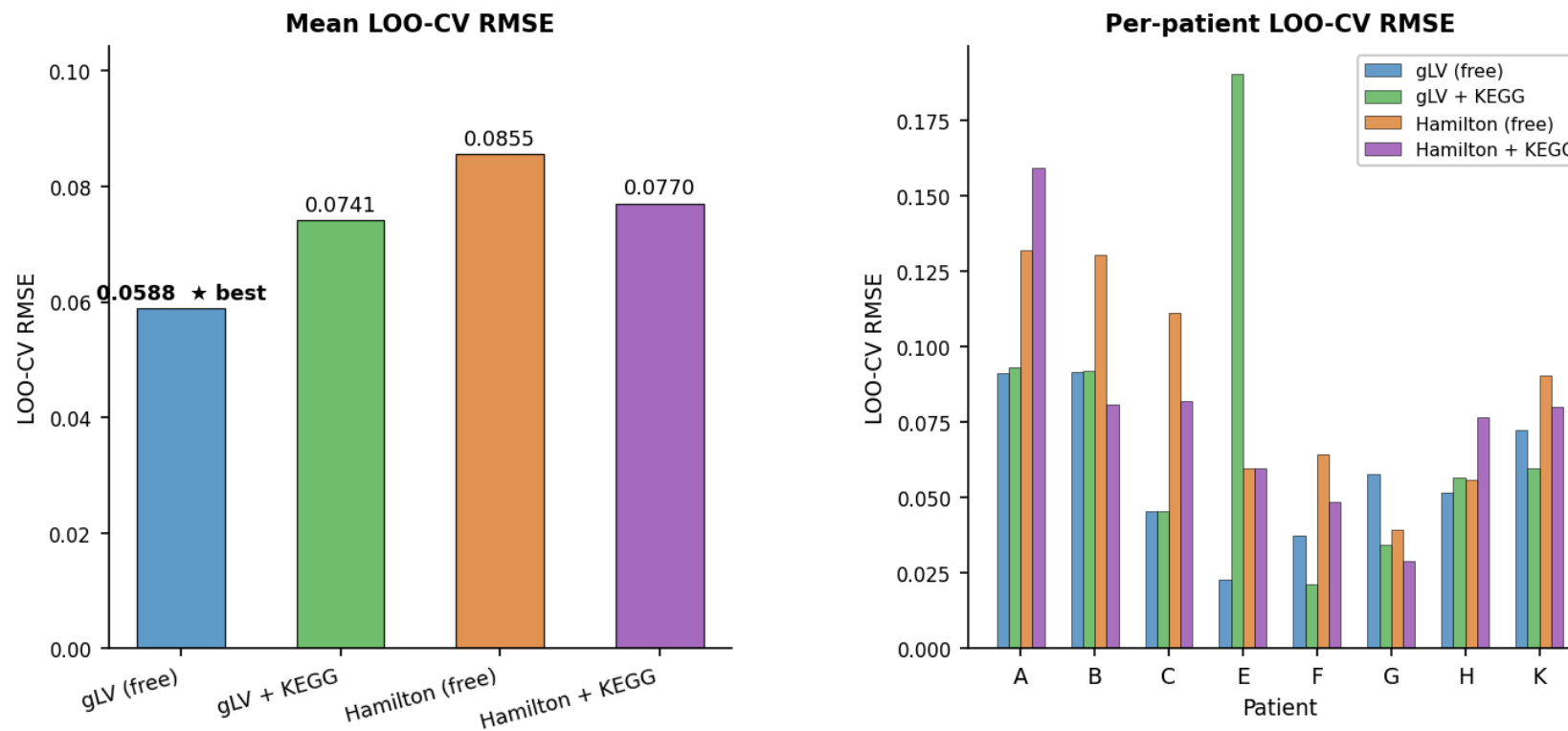
r — best patient (G)

0.148

r — worst patient (K)

Result 3 — Model Comparison (LOO-CV)

Guild model comparison: gLV vs Hamilton × free prior vs KEGG prior



Hamilton + KEGG LOO-CV = 0.0770 · 9.4% improvement over Hamilton free (0.0855) · Competitive with gLV + KEGG (0.0741, $\Delta = 3.9\%$)

Key Findings

What works ✓

- **Full sign consistency:** 22/22 (100%) KEGG agreement — metabolomic constraints are compatible with Hamilton dynamics
- **KEGG prior regularises effectively:** LOO-CV 0.0855 → 0.0770 (9.4% improvement)
- **Competitive generalisation:** within 3.9% of gLV + KEGG at 0.0741
- **Biologically interpretable $\mathcal{A}$:** negative self-regulation, positive Actinobacteria ↔ Bacilli mutualism
- **Steady-state check:** RMSE 0.177 (vs ODE 0.083) — data are transient, not equilibrium

Remaining gap vs gLV

- gLV free still best at **0.0588** — Hamilton's transient approximation (short t) limits fit quality
- Patient A outlier (LOO-RMSE 0.159) drives variance; community type differs from training set
- **Next:** expanded flow (34 pairs, all 10 patients) fit running → may close the gap

Open questions

- Do **34 KEGG+AGORA pairs** further improve LOO-CV?
- Optimal σ is **insensitive** ($\sigma \in [0.05, 0.30]$ all give SA = 22/22, RMSE ≈ 0.081)

Next Steps

#	Task	Status
1	Hamilton + KEGG LOO-CV (8-fold)	✓ Done — 0.0770
2	Aggregate fold results & regenerate Fig 3	✓ Done
3	Sign probability heatmap from 10,000p posterior	✓ Done
4	Paper S1 update with correct posterior statistics	✓ Done
5	Expanded flow (34 pairs, full 10 patients) fit	🔄 Running GPU
6	σ sensitivity analysis	✓ Insensitive — $\sigma = 0.15$ optimal
7	Steady-state reformulation	✓ Tested — transient data, ODE preferred
8	Network visualisation of interaction matrix	✓ Done

Summary: Hamilton + KEGG prior achieves 100% metabolomic sign consistency (22/22) and 9.4% LOO-CV improvement over Hamilton free (0.077 vs 0.086). Generalisation is competitive with gLV + KEGG ($\Delta = 3.9\%$), while offering mechanistic interpretability that gLV lacks.